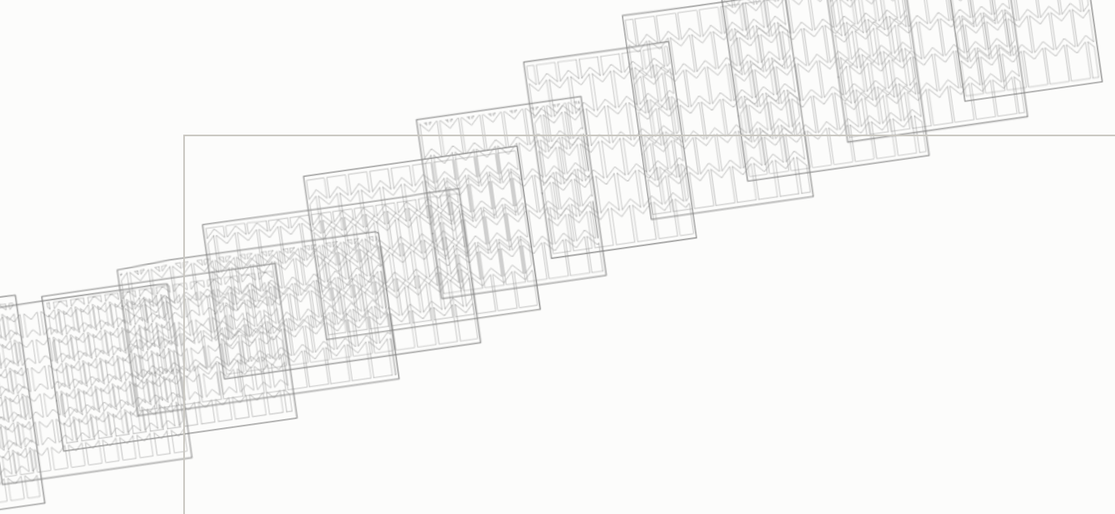
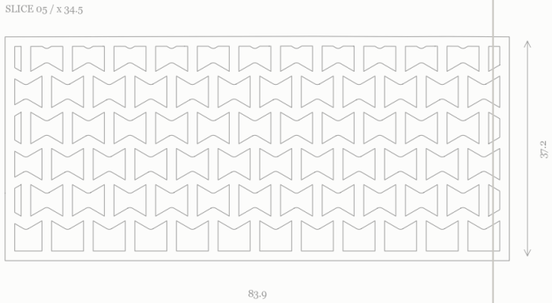
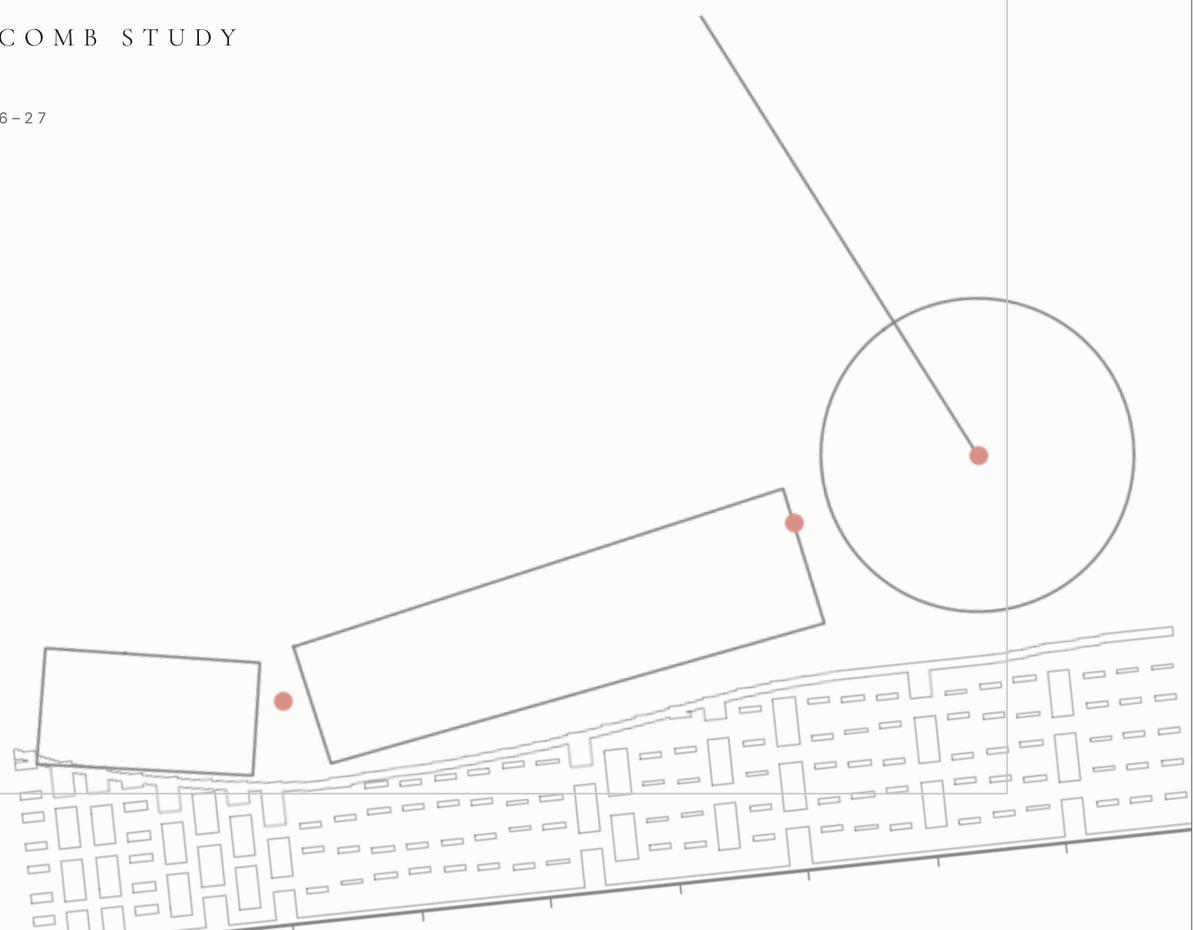




AUXETIC MIDSOLE

A RE-ENTRANT HONEYCOMB STUDY

MOS PROJECT · 2026–27



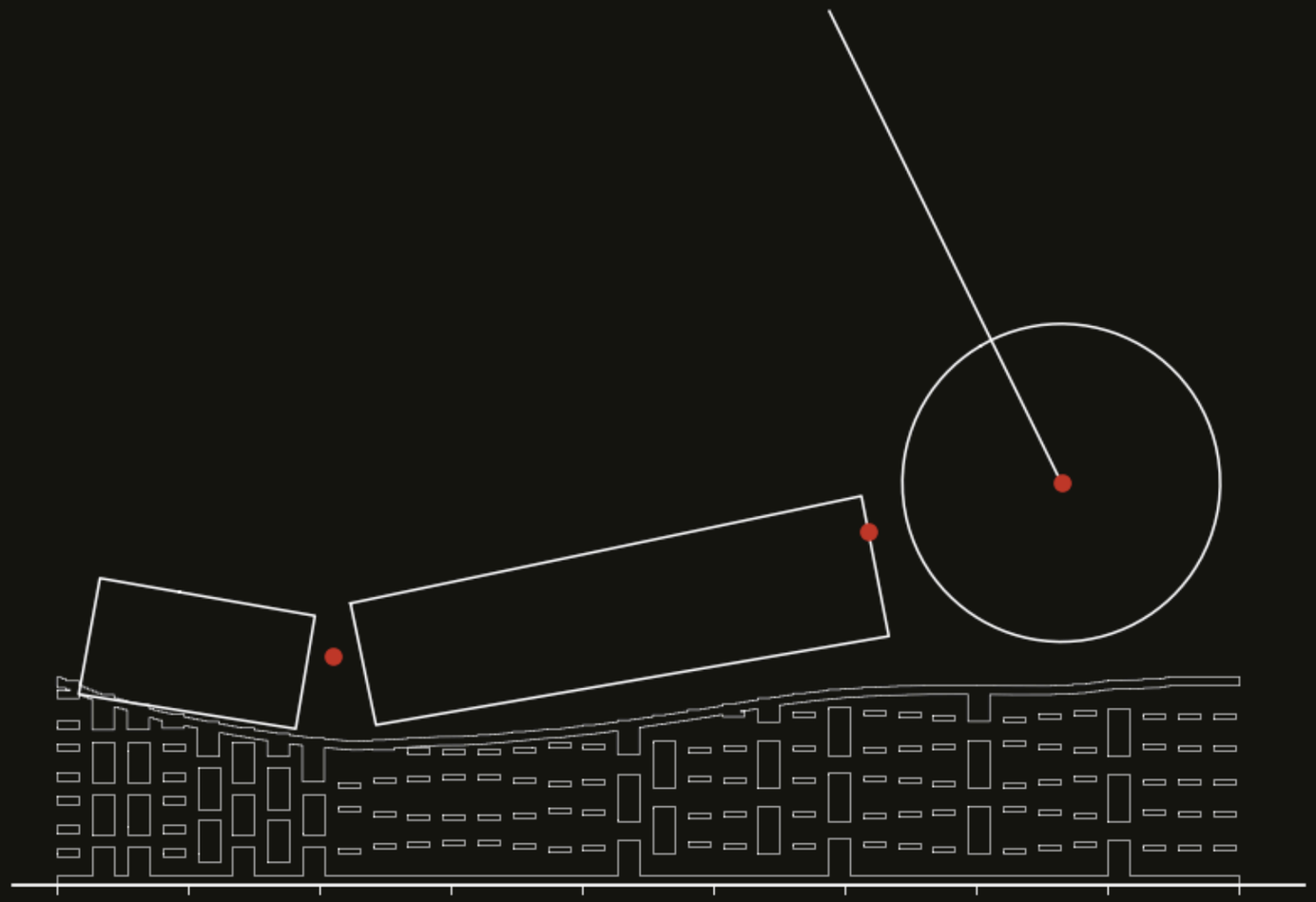
I

A SOLE THAT HUGS BACK

Re-entrant cells contract sideways when squeezed

A conventional foam bulges outward under the foot. A re-entrant honeycomb does the opposite: its bow-tie cells fold inward, gathering material under the load. The midsole is cut into 34 thin slices across the foot, each filled with this lattice, and joined by two plates.

The question this study follows: how does each slice deform under a real running stance, and which cell should go where?



01 Pseudo-foot at mid-stance over the midsole section, y = 60 mm. Red points: ankle, heel–midfoot and MTP hinges.

II

FOUR WAYS TO READ A CELL

Gibson & Ashby · Masters & Evans · Malek & Gibson · Ohtaki

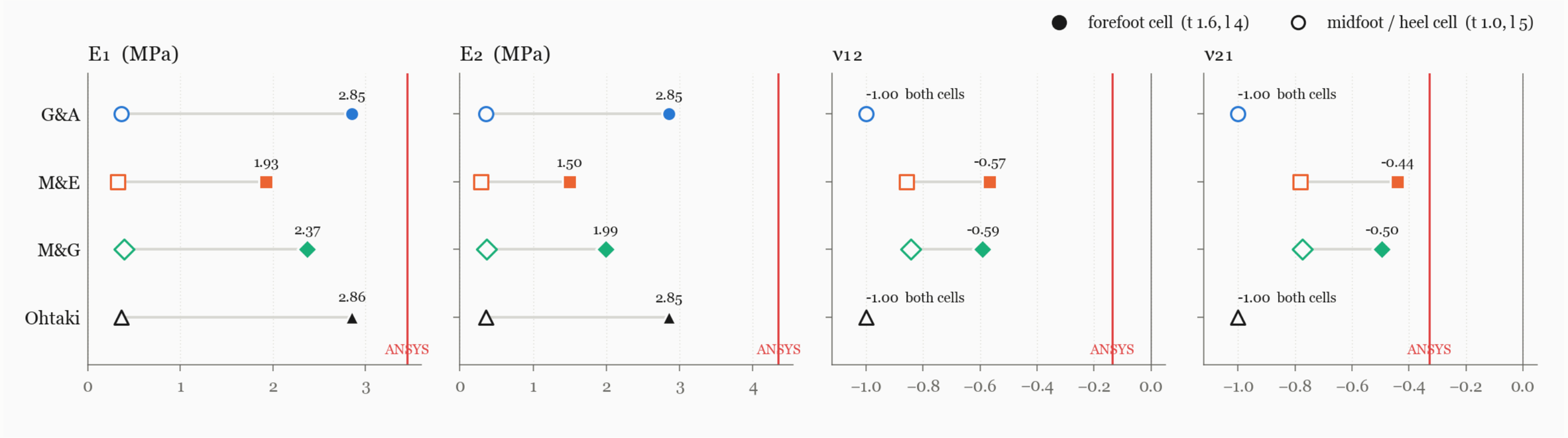
① wall compliances, in units of $l^3 / E_s\, b\, t^3$

$C_p = 1 + r_h \quad C_a = (t/l)^2 \quad C_v = 2\,(h/l)(t/l)^2$

② $E2 = E_s\,(t/l)^3\,(h/l + \sin\theta) \,/\, (\cos\theta \cdot D2)$

$D2 = \cos^2\theta\, C_p + \sin^2\theta\, C_a + C_v$

③ Ohtaki: each half wall an elastica, $\varphi'' = f_x \sin\varphi - f_y \cos\varphi$



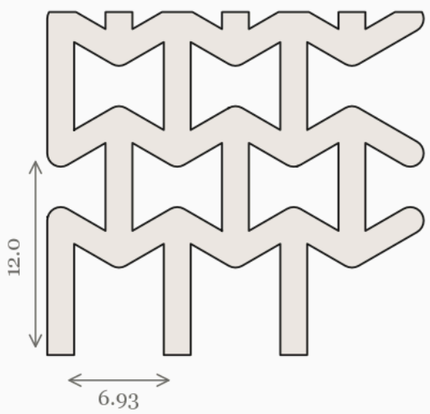
02 G&A keeps bending only ($C_a = C_v = 0$); M&E adds stretching; M&G shortens the bending length at rigid joints; Ohtaki lets the walls rotate freely.

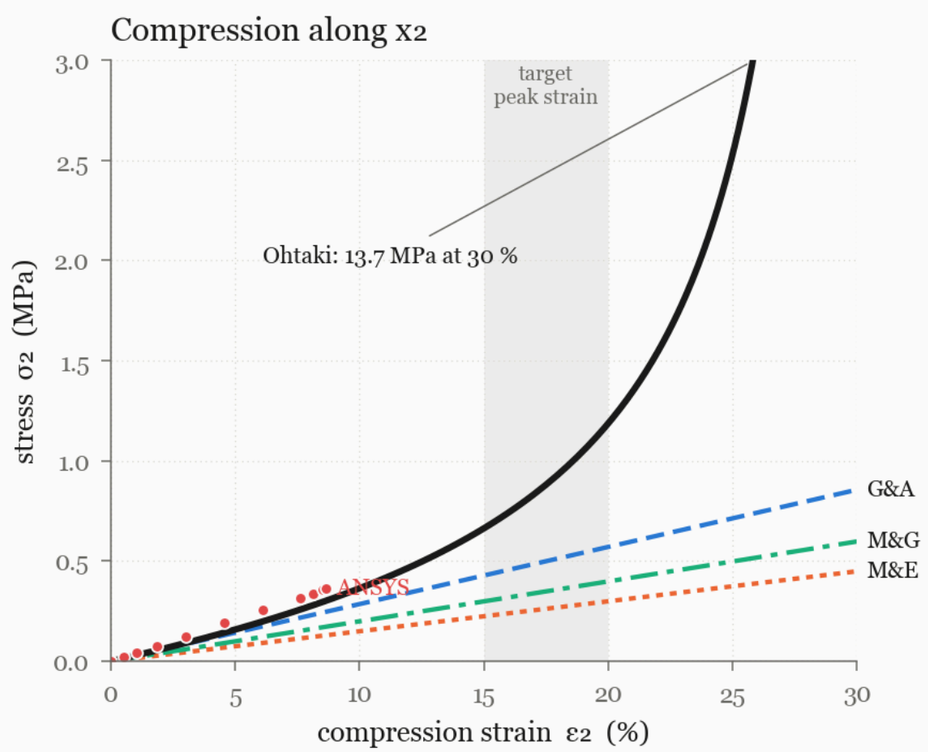
III

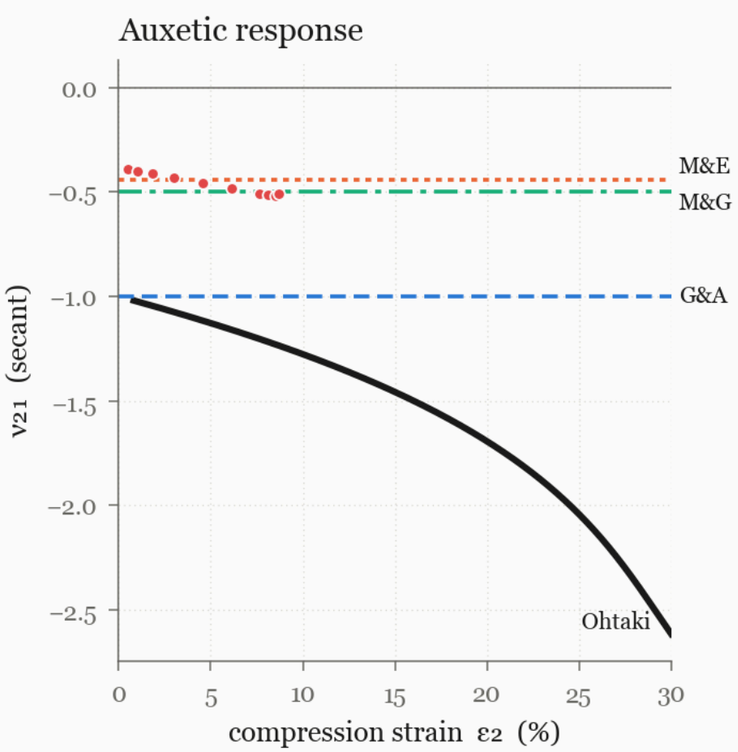
UNDER THE FOREFOOT

Small strain agrees; large strain is where the models part

θ -30° l 4 h 8 t 1.6 mm







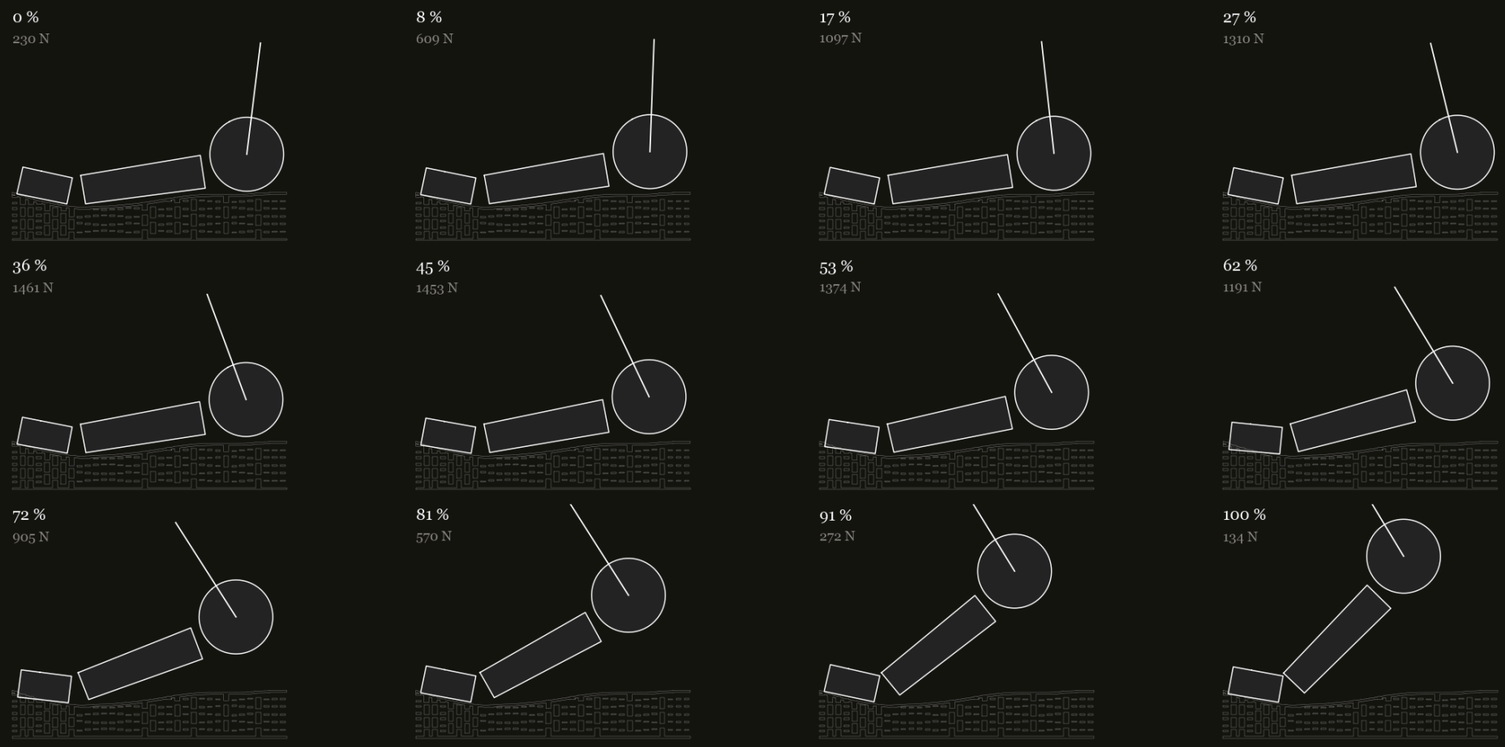
03 Forefoot cell θ −30°, l 4, h 8, t 1.6 mm. ANSYS plane-stress results (4 × 4 cells, 5 mm thick) are overlaid in red once run.

IV

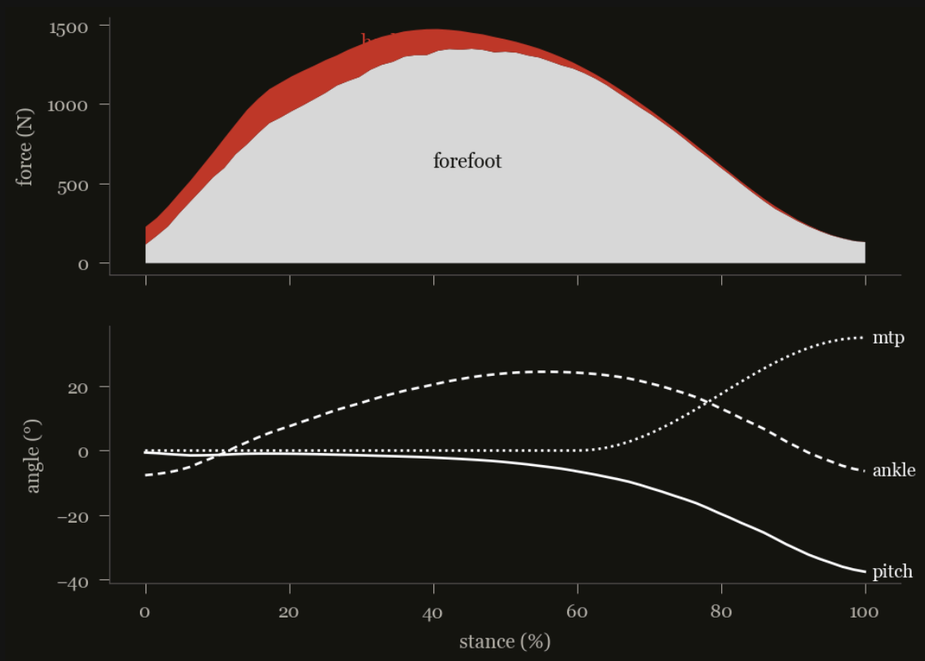
213 MILLISECONDS

One right-foot stance at 3.5 m/s, Fukuchi et al. dataset

Force and centre of pressure give the load and its heel-to-toe split; markers give the foot pitch and ankle angle. The toe bend follows Stefanyshyn & Nigg. This runner lands on the forefoot: the heel barely carries load.



04 Twelve frames of stance: pseudo-foot posed from the measured angles, heel and midfoot rigid, toes bending about the MTP.



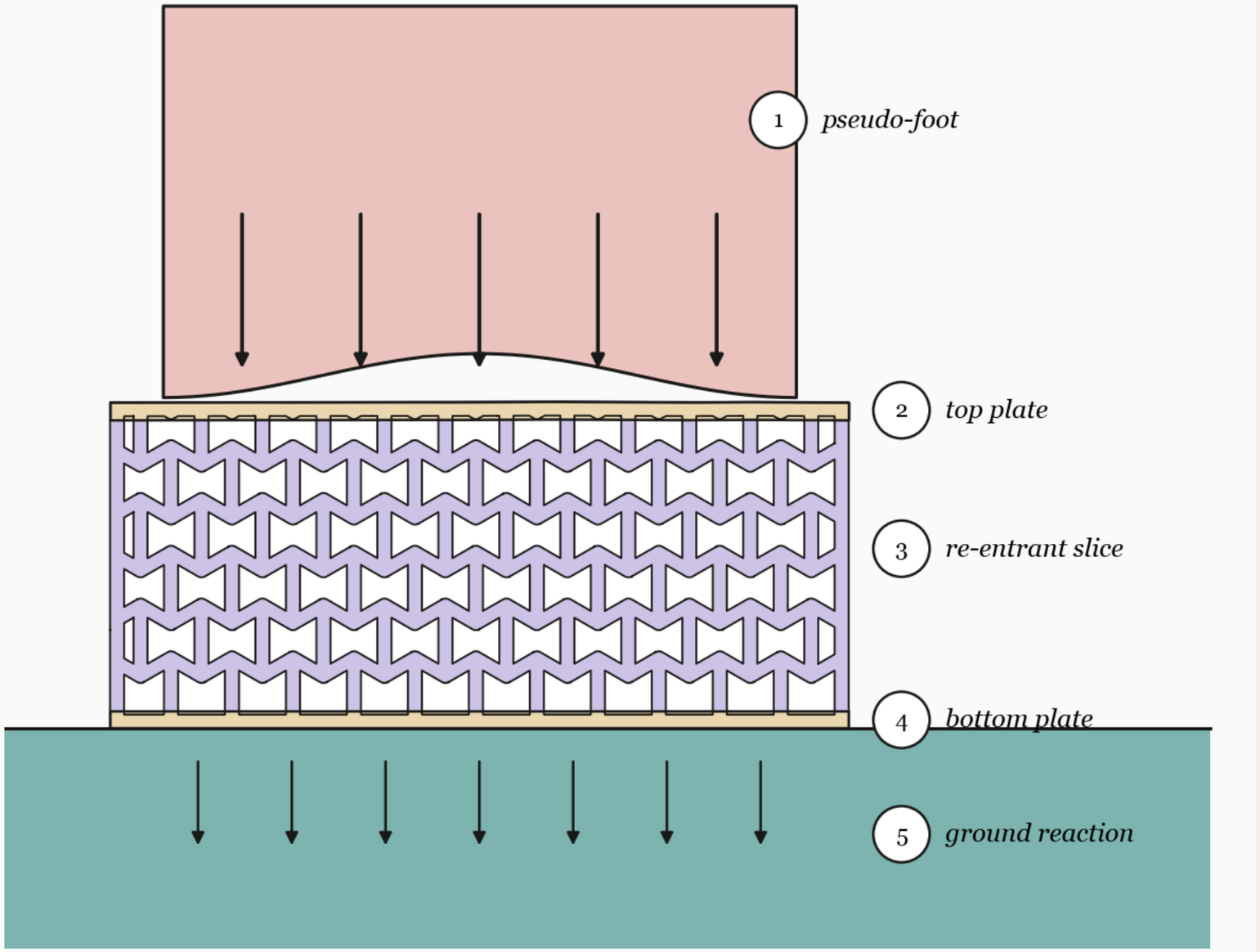
05 Force split (white forefoot, red heel) and joint angles, smoothed by a zero-lag moving average.

V

LOAD PATH

Each slice is a beam on a lattice foundation

1. The posed foot sinks until the slices carry the measured force.
2. The top plate bends across the foot, spreading the load.
3. Every lattice column answers with Ohtaki's large-strain stress.
4. The bottom plate ties the slices together.
5. The ground returns F, row by row through the stance.



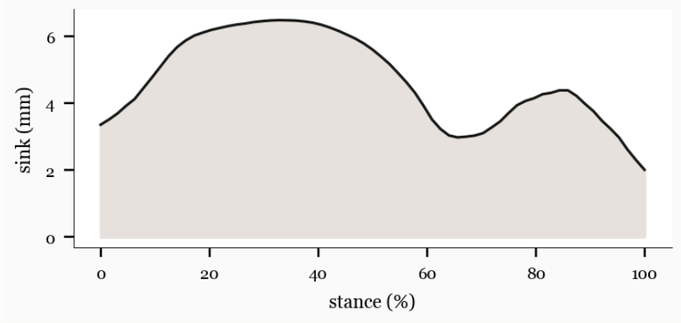
06 Cut-away across the foot at x = 34.5 mm (forefoot slice).

VI

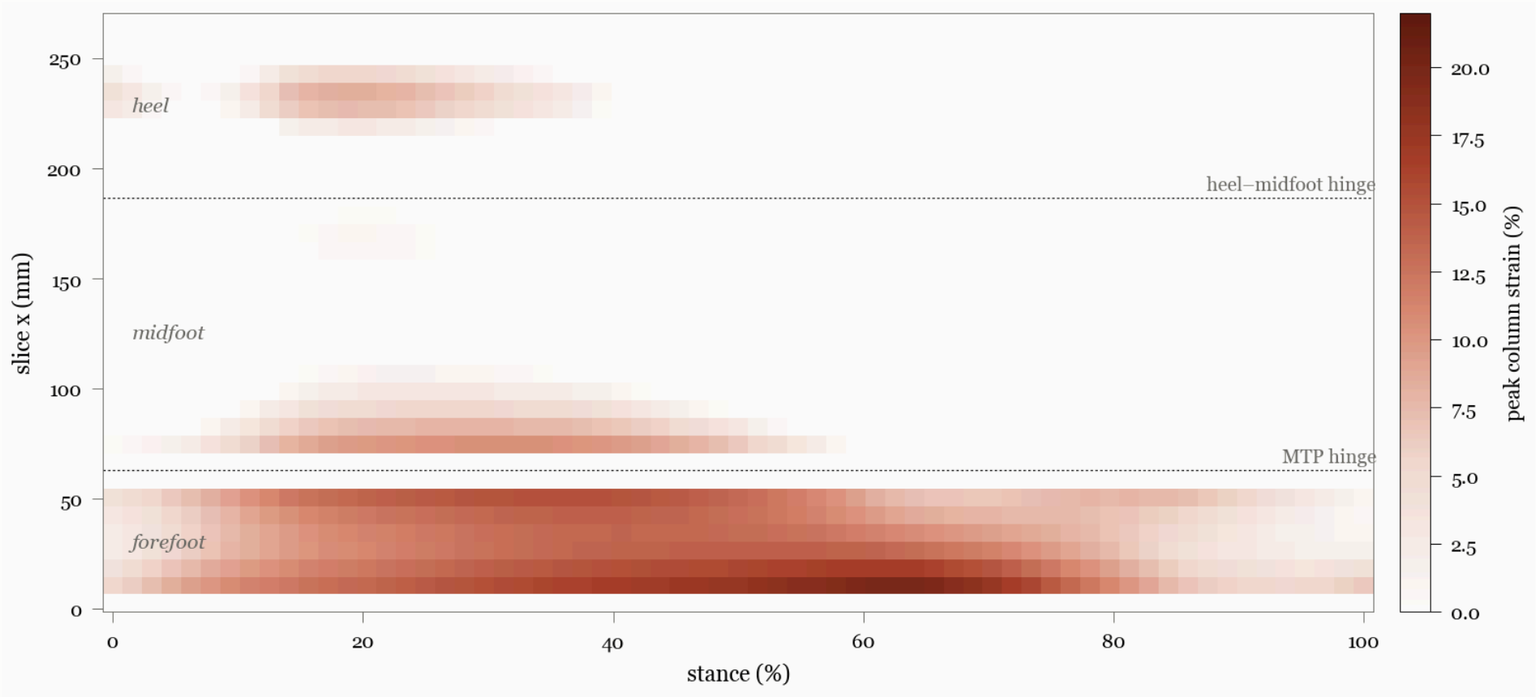
WHERE THE SOLE WORKS

Peak lattice strain, slice by slice, through the stance

For every frame the sink is solved so the slice forces add up to the measured load. Strain gathers under the metatarsal heads and fades toward the heel.



08 Sink of the foot into the sole, solved per frame.



07 Zoned lattice. Dashed: the hinge lines that set the zone limits.

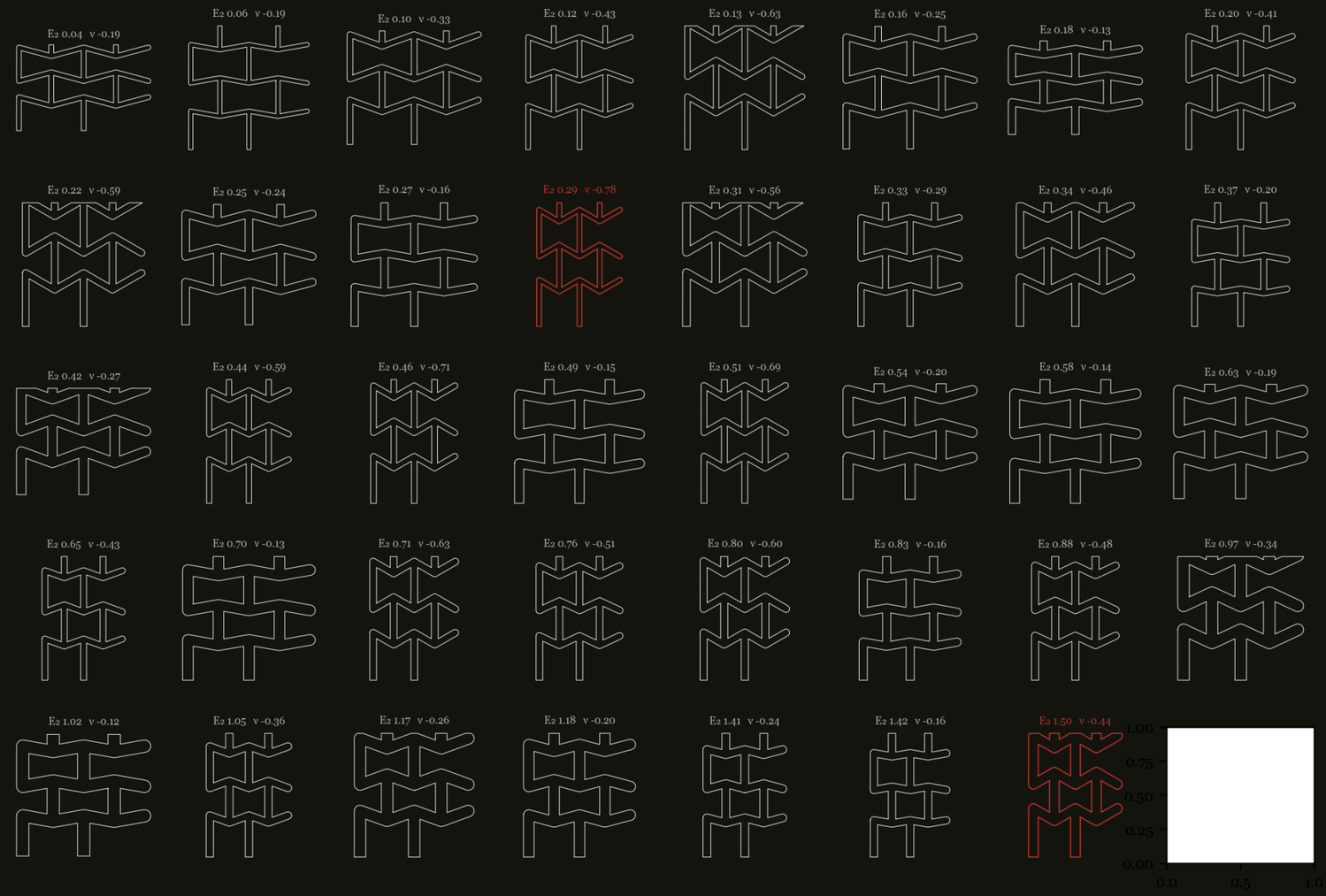
VII

TWO HUNDRED CELLS

Searched per zone for a 15–20 % peak strain

zone	θ	l	h	t	E_2 MPa	ν_{21}	peak
forefoot	−30°	4	8	1.6	1.50	−0.44	19.8 %
midfoot	−30°	5	10	1.0	0.29	−0.78	10.5 %
heel	−30°	5	10	1.0	0.29	−0.78	8.4 %

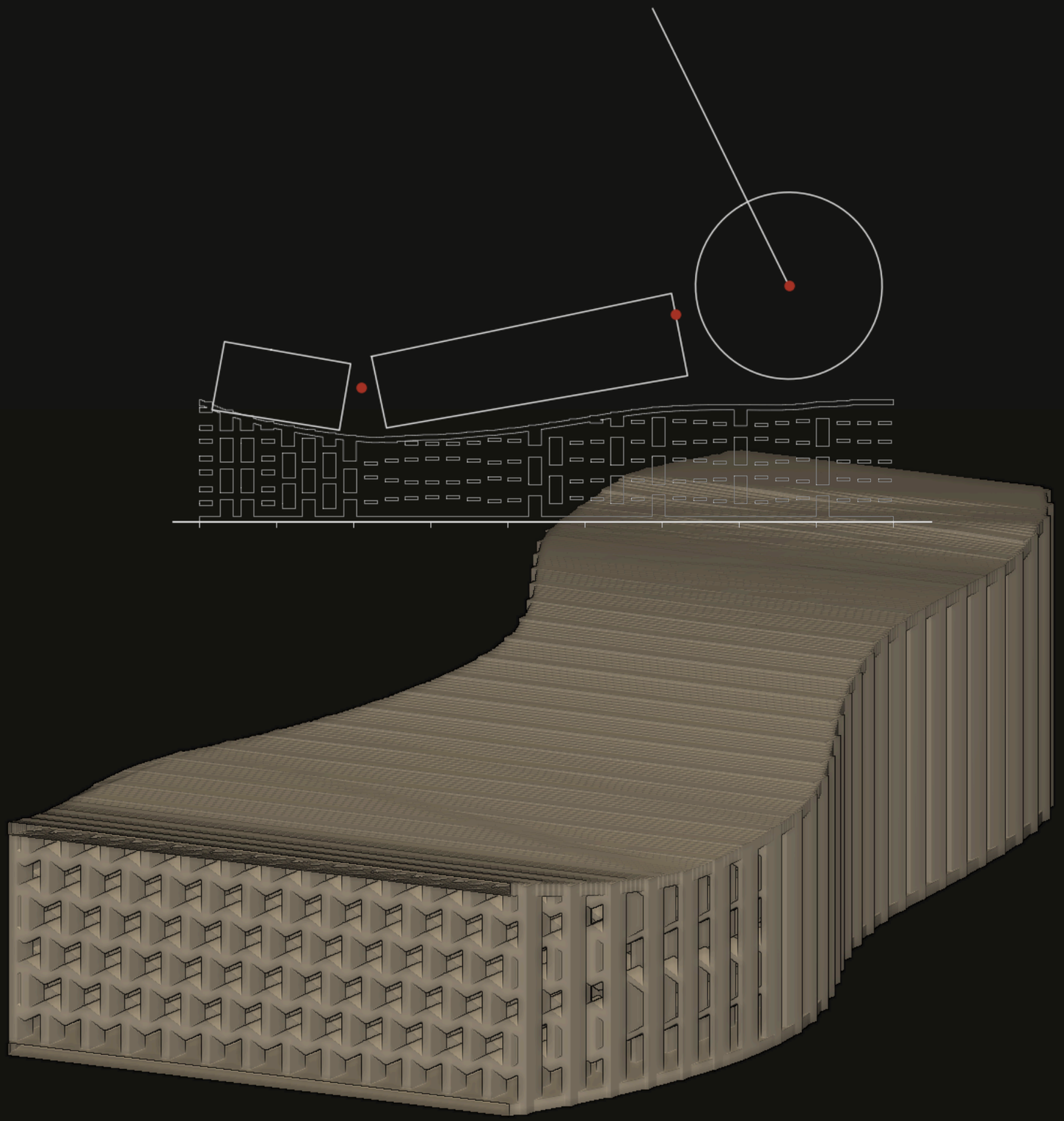
Midfoot and heel strain is set by the foot's shape, not its force, so the search settles on the softest, most auxetic cell allowed.



09 Forty of the candidates, softest to stiffest (M&E). Red: chosen.

VIII

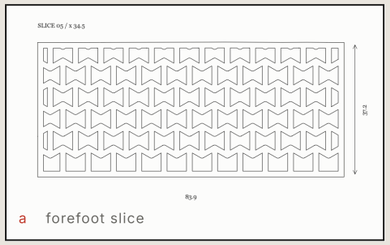
SECTION



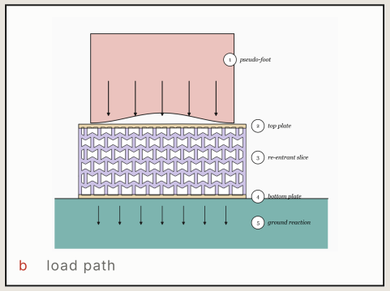
10 Zoned midsole cut across the foot at x = 38 mm, through the forefoot lattice. Above: the stance elevation.

IX

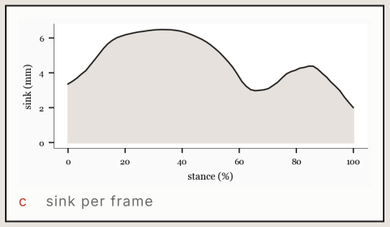
ASSEMBLY



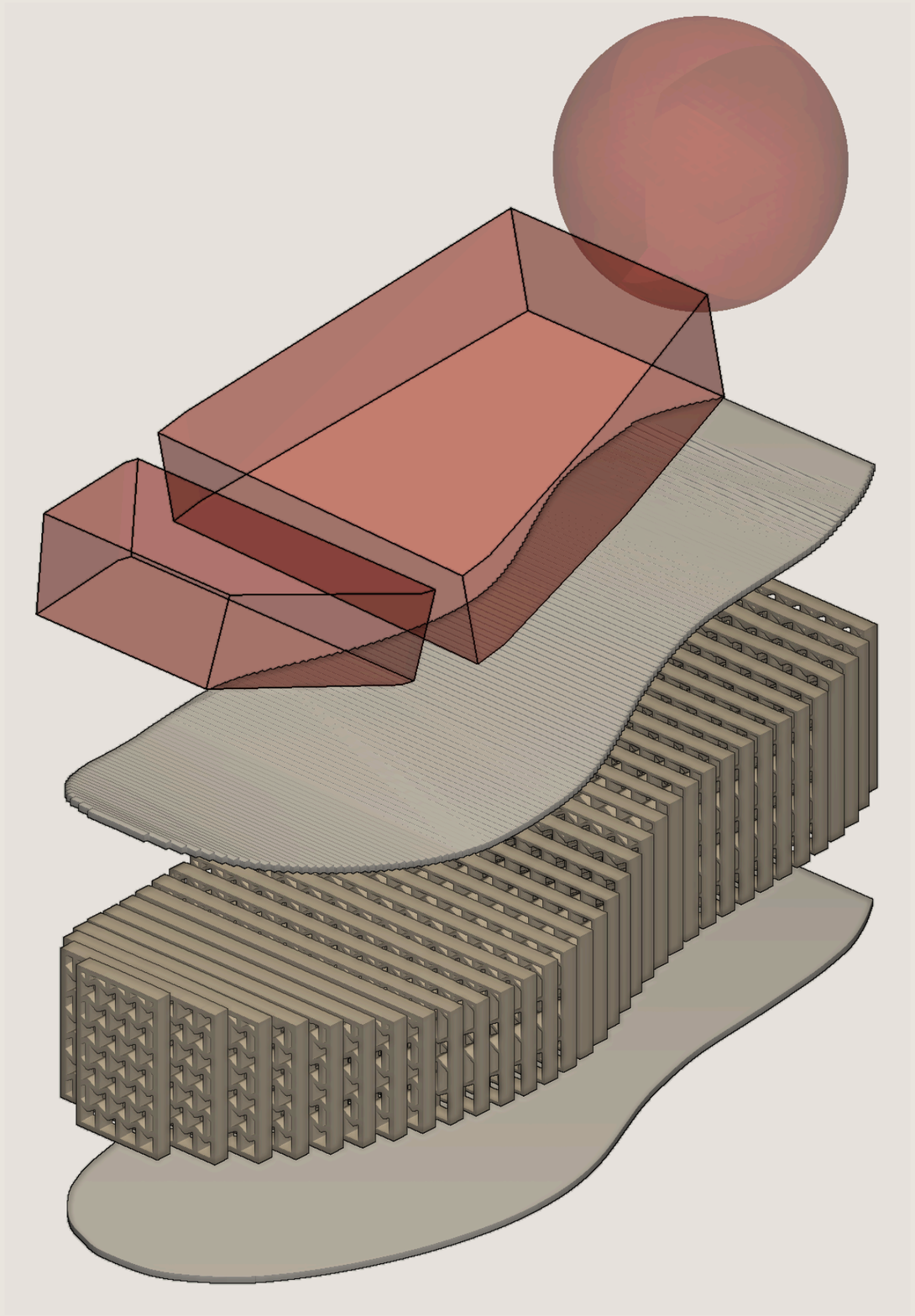
a forefoot slice



b load path



c sink per frame



11 Top to bottom: pseudo-foot (red), top plate, 34 zoned slices, bottom plate. 242 cm³, one watertight body.

X

WHAT IS NOT YET KNOWN

1. One runner, a forefoot striker: a heel striker needs a stiffer heel.
2. t/l up to 0.4 strains slender-beam theory; ANSYS decides.
3. Ohtaki ignores walls touching; trust it to about 20 %.
4. The heel–midfoot joint is rigid; no midfoot marker exists.
5. Slices only interact through the plates (Winkler, no shear).

Next: three dimensions

A strut lattice that is re-entrant along every axis would remove the gaps, carry the forefoot flex itself and dome under the foot instead of saddling. Its stiffness scales with density squared, so the material question remains.

